

JMM Proposal Outline

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1 Title: Applications of Topology

2 Name, Affiliation, Email

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3 Public Description: 500 Characters

Topology is easy to think of as a discipline of math with little relevance to the “real” world; however, the broad idea of finding “shape” in data has led to novel insights in various fields, including neuroscience and cancer research. In this session, speakers will discuss their use of topological ideas and frameworks in their work, as well as discuss mathematical tools and theory used in topological data analysis.

4 Sample Speaker List

- Brad Theilman, Sandia National Laboratories
- Enrique Alvarado, Iowa State University

5 Description: 3000 Characters

This session will bring together speakers from many disciplines, under the umbrella of topological data analysis (TDA) and visualization. This will include mathematicians who study algorithms such as Mapper and concepts like persistent homology and dimension reduction, as well as researchers who are not mathematicians who have used these ideas successfully in their work. There will be ample discussion of software and opportunities to collaborate on larger

TDA and visualization libraries, an area currently lacking in availability, user friendliness, and standardization.

There has been a variety of applications of Mapper to data in various domains (such as: neuroscience, clinical medicine, genomics, and sociology) each with their own approach to parameter selection and analysis. Despite this growing body of work, the field lacks a standardized methodology for parameter tuning. Additionally, the software landscape for Mapper is fragmented: implementations differ in their supported data formats, scalability, library APIs, and visualization capabilities. This leaves researchers unaware of best practices or critical technical details from other fields, which can lead to suboptimal, misleading, or even invalid analyses.

6 MSC

One of:

- 101: Teaching and Learning
- 102: Recreational Mathematics
- 103: Professional Development and Professional Concerns
- 104: Wider Issues